# Calder Lenhart

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## EDUCATION

2025–2027 **M.S. Physics**; Ohio State University
GPA: 3.72/4
Coursework funded by the Faculty and Staff Tuition Assistance Plan – all courses taken for credit, concurrently with my full-time employment in the Department of Astronomy

Graduate-level coursework: Elementary Particle Physics, Differential Geometry (Math Dept.), Quantum Mechanics 1-2, Electromagnetic Field Theory, Analytic and Numeric Methods of Physics, Classical & Statistical Physics I

2020–2024 **B.S. Physics, Astronomy & Astrophysics**; Ohio State University
Minor in Mathematics, Minor in History
Magna cum laude, with Research Distinction in Astronomy & Astrophysics
Thesis: Phase-Resolved Cross-Correlation Transmission Spectroscopy of KELT-20b

Coursework: Applied Differential Geometry, Cosmology, Honors Quantum Mechanics I-II, Honors Intermediate E&M I-II, Statistical Mechanics, Real Analysis, Big Data Analytics in Physics

## PUBLICATIONS

### Refereed Publications

[1] **C. Lenhart**, M. C. Johnson, J. Wang, A. P. Asnodkar, S. Petz, A. Duck, K. G. Strassmeier, and I. Ilyin. "PEPSI Investigation, Retrieval, and Atlas of Numerous Giant Atmospheres (PIRANGA). II. Phase-resolved Cross-correlation Transmission Spectroscopy of KELT-20b". In: *The Astronomical Journal* 171.2 (Jan. 2026), p. 81. DOI: 10.3847/1538-3881/ae2252.

[2] C. Basinger, M. C. Johnson, J. Wang, A. Duck, A. Pai Asnodkar, S. Petz, **C. Lenhart**, I. Ilyin, and K. Strassmeier. "PEPSI investigation, retrieval, and atlas of numerous giant atmospheres (PIRANGA) – III. Composition and winds in the atmosphere of TOI-1518 b". In: *Monthly Notices of the Royal Astronomical Society* 543.4 (Sept. 2025), pp. 4136–4143. DOI: 10.1093/mnras/staf1648.

[3] V. Bonidie, M. C. Johnson, J. Wang, S. Petz, J. Kamen, **C. Lenhart**, A. Duck, C. Badenes, K. Strassmeier, and I. Ilyin. "PEPSI Investigation, Retrieval, and Atlas of Numerous Giant Atmospheres (PIRANGA). IV. High-resolution Phased-resolved Spectroscopy of the Ultra-hot-Jupiter KELT-20 b". In: *The Astronomical Journal* 171.1 (Dec. 2025), p. 34. DOI: 10.3847/1538-3881/ae21be.

### Manuscripts in Preparation

[1] **C. Lenhart** and C. S. Kochanek. "A Systematic Search for Dippers, Long-Term Variables, and Microlensing Events in ASAS-SN". Manuscript in preparation. 2026.

[2] **C. Lenhart** and T. A. Thompson. "When Inside Out is Outside In: Fountain Flows Launched from Circumnuclear Starbursts Drive Accretion, Star Formation, and Enrichment in Outer Disks". Manuscript in preparation. 2026.

### Conference Abstracts

[1] C. Basinger, M. Johnson, J. Wang, A. Duck, A. Pai Asnodkar, S. Petz, **C. Lenhart**, and K. Strassmeier. "Composition and Winds in the Atmosphere of TOI-1518 b". In: *American Astronomical Society Meeting Abstracts*. Vol. 247. American Astronomical Society Meeting Abstracts. Feb. 2026, 443.69, p. 443.69.

[2] J. Kamen, M. Johnson, V. Bonidie, S. Petz, **C. Lenhart**, J. Wang, and K. Strassmeier. "Chemistry and dynamics from high-resolution emission spectra of the ultra-hot Jupiters KELT-20b and TOI-1518b from the PIRANGA project". In: *American Astronomical Society Meeting Abstracts*. Vol. 247. American Astronomical Society Meeting Abstracts. Feb. 2026, 114.06, p. 114.06.

[3] **C. Lenhart**, M. Johnson, S. Petz, J. Wang, A. Pai Asnodkar, K. Strassmeier, and I. Ilyin. "Analysis of KELT-20b's Atmospheric Dynamics Using PEPSI: Line Profiles During Transit and Velocity Offsets". In: *American Astronomical Society Meeting Abstracts*. Vol. 243. American Astronomical Society Meeting Abstracts. Feb. 2024, 179.09, p. 179.09.

### Theses

[1] **C. Lenhart**. "Phase-Resolved Cross-Correlation Transmission Spectroscopy of KELT-20b". Undergraduate Thesis. The Ohio State University, Dec. 2024.

### TNS Discovery Reports

[1] K. Z. Stanek and **C. Lenhart**. *ASAS-SN Transient Discovery Report for AT 2026lru / ASASSN-26df*. Transient Name Server. May 8, 2026.

[2] K. Z. Stanek and **C. Lenhart**. *ASAS-SN Transient Discovery Report for AT 2026osq / ASASSN-26dr*. Transient Name Server. June 9, 2026.

## PRESENTATIONS

### Talks

[1] **C. Lenhart** and M. C. Johnson. *Analysis of an Ultra-Hot Jupiter's Atmosphere*. Ohio State Department of Astronomy Summer Undergraduate Research Program Symposium (Columbus, OH, USA). July 2023.

### Posters

[1] **C. Lenhart**, M. C. Johnson, J. Wang, A. P. Asnodkar, S. Petz, K. G. Strassmeier, and I. Ilyin. *Analysis of KELT-20b's Atmospheric Dynamics Using PEPSI: Line Profiles During Transit and Velocity Offsets*. 243rd Meeting of the American Astronomical Society (New Orleans, LA, USA). Jan. 2024.

## AWARDS & GRANTS

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| 2026 | Honorable Mention, NSF Graduate Research Fellowship Program, Astronomy & Astrophysics |
| 2025 | Artist Grant, Greater Columbus Arts Council |
| 2024 | Ann Slusher Tuttle Undergraduate Scholarship, Ohio State University Department of Astronomy |
| 2023 | Undergraduate Research Scholarship, Ohio State University College of Arts and Sciences |
| 2023 | 1st Place, MakeOHI/O 2023 |
| 2020 | Eagle Scout, Boy Scouts of America |
| 2019 | Youngstown CityScape Beautification Watch Award |

## RESEARCH EXPERIENCE

2026 – **Post-baccalaureate researcher, Theoretical Astrophysics**, Department of Astronomy, Ohio State University
(Advisor: Todd Thompson)

- Developing analytic and numerical models of galactic fountains launched by circumnuclear starbursts to investigate how angular momentum-deficient gas recycling drives radial accretion, sustains disk star formation, and shapes metallicity gradients.
- Derived coupled steady-state conservation equations for mass, angular momentum, and metals, incorporating observed rotation curves, gas surface densities, star formation rates, and H II region abundances to model radial inflows and chemical enrichment in six nearby spiral galaxies.
- Modeled fountain interactions with the circumgalactic medium, incorporating mass entrainment, angular momentum exchange, and metal mixing to derive observational constraints on the composition and kinematics of accreting gas.
- Investigating magnetized proto-neutron-star winds using magnetohydrodynamic simulations with `PLUTO`, exploring the effects of magnetic field strength, rotation, and neutrino heating on wind dynamics, mass loss, and unbound outflows.
- Developing GPU-accelerated magnetohydrodynamic simulations with `gPLUTO` to investigate proto-neutron-star outflows across different magnetic-field strengths and initial conditions.

2025– **Post-baccalaureate researcher, Time-Domain Astronomy**, Department of Astronomy, Ohio State University
(Advisors: Christopher Kochanek, Krzysztof Stanek)

- Built a parallelized processing pipeline for $> 17$ million ASAS-SN light curves to search for transient brightenings from gravitational microlensing, irregular dimmings from circumstellar dust obscuration, and long-term or periodic variability from magnetic activity, rotation, and starspots.
- Used Gaussian Process models to separate quiescent stellar and systematic variability from anomalous brightness changes.
- Built a Bayesian event detection framework using marginalized log-likelihood grids, Bayes factors, and posterior probabilities to identify statistically significant dimmings and brightenings relative to each source's quiescent baseline.
- Built injection-recovery tests by injecting synthetic dip and jump morphologies into real light curves to measure detection efficiency as a function of event depth, event duration, baseline magnitude.
- Used catalog cross-matches, infrared colors, dust-extinction corrections, and a `LightGBM` model trained on 107 features to classify candidates as microlensing events, dusty dimming events, flares, YSOs, CVs, starspot variables, or other variable classes.

2025– **Post-baccalaureate researcher, Exoplanet Atmospheres**, Department of Astronomy, Ohio State University
(Advisor: Marshall C. Johnson, Co-advisor: Ji Wang)

- Building a GPU-accelerated Bayesian atmospheric retrieval pipeline in Python using `JAX`, `NumPyro`, and `ExoJAX`, with SVI and HMC-NUTS posterior inference.
- Implemented multiple atmospheric chemistry models, including free-retrieval abundances, altitude-dependent abundances, FastChem thermochemical equilibrium, and hybrid equilibrium/free-chemistry models.
- Performing Bayesian retrievals of ultra-hot Jupiter atmospheres, including KELT-9b, KELT-20b, TOI-1431b, and more, using high-resolution PEPSI/LBT spectroscopy.
- Constraining temperature-pressure profiles, atomic and molecular abundances, C/O ratio, metallicity, and atmospheric velocity structure in transmission and emission geometries.
- Developing joint retrievals combining high-resolution spectroscopy with public low-resolution optical/infrared spectra and broadband photometry to probe atmospheric structure across multiple pressure levels.

2023–2024 **Undergraduate Researcher, Exoplanet Atmospheres**, Department of Astronomy, Ohio State University
(Advisor: Marshall C. Johnson, Co-advisor: Ji Wang)

- Led a first-author study using PEPSI/LBT spectroscopic time-series observations to extract Doppler-shifted atmospheric absorption from thousands of weak lines in the ultra-hot Jupiter KELT-20b.
- Built a high-resolution spectroscopy processing pipeline to remove telluric and stellar contamination, inject and recover model signals, and cross-correlate residual spectra with `petitRADTRANS` forward-modeled spectra.
- Detected atmospheric absorption at high significance, including Fe and $Fe^{+}$, and phase-resolved the strongest signals to map atmospheric velocity structure across the transit.
- Identified statistically significant start/end-of-transit asymmetries, supporting altitude-dependent atmospheric structure, ion-neutral differences, and the scale-height effect.
- Showed that adopted literature system parameters can bias measured atmospheric velocities; proposed a standardized measurement procedure that improved consistency across observations.

2022–2023 **Undergraduate Research Assistant, Materials Science**, Department of Materials Science and Engineering, Ohio State University
(Advisor: Prof. Sheikh Akbar)

- Performed hydrothermal synthesis of metal-oxide nanostructures for gas sensors; characterized resistivity, response time, and selectivity of analytes.
- Cleanroom trained at Nanotech West Lab; managed safe handling of hazardous gases and materials.

## WORK EXPERIENCE

2025– **Research Technician I**, The Ohio State University, On-site
Prof. Christopher Kochanek, Prof. Krzysztof Stanek (Department of Astronomy)

- Perform daily quality control on images from 20 cameras on 5 ASAS-SN telescopes; remotely recalibrate telescopes; verify supernova and other transient candidates; release confirmed discoveries to the public.

2025 **Physics AI Trainer**, Mercor Intelligence

- Trained a top AI research organization's LLM on undergraduate and graduate-level physics and math problems.

2022–2023 **Private Tutor**, Grade Potential Tutoring

- Tutored middle school, high school, and college students in-person in STEM.

2022–2023 **Private Tutor**, Wyzant

- Maintained a 5.0/5.0 rating across 30+ reviews and earned recognition as a top tutor in Columbus, OH.
- Created customized study plans for students ranging from middle school to college level in math, physics, and standardized test prep.

2021–2022 **Math Tutor**, Ohio State University Mathematics and Statistics Learning Center

- Tutored Calculus I–III students, managing scheduling, lesson planning, and teaching of more than 20 students per week.

## PROJECTS

2026– **`DustyCult`**, GitHub
- Writing Julia tools for modeling transient stellar dimming from circumstellar dust occultation, including dust forward models, likelihoods, priors, and Turing sampling support.

2026– **`mlexo`**, GitHub
- Developing MLX-based astronomical time-series and light-curve modeling tools.

2026– **`atmo-retrieval`**, GitHub
- Building a Bayesian exoplanet atmosphere retrieval pipeline using `ExoJAX`, `NumPyro`, and `PySysRem` for high-resolution exoplanet spectra.

2026– **`exoJAX`**, GitHub
- Contributed to the auto-differentiable high-resolution spectrum modeling code for exoplanets/brown dwarfs using JAX. This code is the backend for `atmo-retrieval`.

2025– **`malca`**, GitHub
- Built `MALCA`, a Bayesian multi-timescale ASAS-SN light-curve analysis pipeline for detecting dimming/dipping events, microlensing candidates, and long-term variability using Gaussian-process baselines, marginal-likelihood scoring, multi-stage filtering, and multi-wavelength characterization.

2023–2025 **`atmo-analysis`**, *private GitHub repository*
- Developed LBT/PEPSI high-resolution exoplanet atmospheric transit analysis code for phase-resolved cross-correlation spectroscopy.

2023 **Linking Writing Processes to Writing Quality**, GitHub
- Developed a Histogram-based Gradient Boosting Regression Tree with Scikit-learn to predict writing quality of mock SAT essays using keystroke logs.
- Engineered features proposed in computational linguistics literature and tuned hyperparameters.
- Placed in the 63rd percentile in Kaggle competition efficiency leaderboard.

2023 **Make OHI/O Makeathon (Intel Challenge)**
- 1st Place Winner: Prototyped an updated cleanroom garment with tear sensors, improved boot covers, and redesigned masks for use in Intel's semiconductor fabrication plants.
- Designed prototype using Arduino boards and authentic cleanroom garment materials.

2021–2022 **Buckeye Solar Racing**, Aerodynamics Team
- Researched solar car geometries and designed canopy/aeroshell components in SolidWorks.
- Ran CFD simulations with STAR-CCM+ and meshed existing canopy with photogrammetry software to validate physical prototypes.

2021 **NASA L'SPACE Mission Concept Academy**, Engineering Team Lead
- Collaborated with a 10-person interdisciplinary team to conceptualize a mission to drill water-ice from the lunar south pole.
- Modeled entry, descent, and landing of a lunar rover; designed a compact lunar regolith drill with SolidWorks; and formulated the CONOPS.
- Performed risk analysis, integration and test, verification and validation.

## SKILLS

**Programming Languages:** Python, C/C++, Julia, Bash/Shell, Linux, Git, HTML, CSS, LaTeX, SolidWorks, Darktable

**Methods:** Exoplanetary atmospheric retrieval, Bayesian inference, MCMC (HMC-NUTS, SVI), Gaussian processes, spectroscopy, photometry, high-performance computing, GPU acceleration, machine learning, Principal Component Analysis

## INTERESTS

**Fine Art Photography:** Recent exhibitions in Columbus, OH include *Exposure*, a two-person show at Wild Gallery in Oct. 2025; *ImageOHIO*, a group show at ROY G BIV in Oct. 2025. Forthcoming three month solo exhibition in Columbus, OH, Nov. 2027 - Jan. 2028.

**Other Activities:** Biking, backpacking, radio show hosting, soccer, tennis, website design